QUALITY INFORMATION DOCUMENT

For
OMI_CLIMATE_TEMPSAL_BALTIC_Ttz_trend

Issue: 4.0

Contributors : P. Lagemaa, U. Raudsepp, I. Maljutenko, V. Huess

Approval Date : November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko	P. Lagemaa
1.1	2019/02/21		Revision	U. Raudsepp I. Maljutenko	P. Lagemaa
2.0	2019/10/10		Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Minor changes	U. Raudsepp I. Maljutenko	P. Lagemaa
3.0	2023/09/01	all	Template change+ minor changes	U. Raudsepp I. Maljutenko	P. Lagemaa
4.0	2025/05/22	all	Template change+ minor changes	P. Lagemaa	V. Huess

TABLE OF CONTENTS

RECORD TABLE **2**

1) INTRODUCTION **4**

a) Product Name 4

b) Indicator Evaluation Method 4

c) Copernicus Marine products used as input for the computation 4

2) SCIENTIFIC VALIDATION **5**

a) Subsurface temperature trend 5

3) REFERENCES **6**

1) INTRODUCTION

a) Product Name

OMI_CLIMATE_TEMPSAL_BALTIC_Ttz_trend.

b) Indicator Evaluation Method

This document provides to Copernicus Marine Service users information on the quality of the horizontally averaged temperature for the Baltic Sea. The linear trends/slopes/fits and their 95% confidence intervals have been calculated using last square regression method described in Draper and Smith (1998) eq. 1.2.9 and 1.4.8.

The daily temperature data is extracted from the BALTICSEA_MULTIYEAR_PHY_003_011 product. The accuracy of the index is directly dependent on the accuracy of the Baltic reanalysis and near-real-time products (BALTICSEA_MULTIYEAR_PHY_003_011) therefore evaluation is based on the QUID documents of the products.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf

2) SCIENTIFIC VALIDATION

a) Subsurface temperature trend

In the CMEMS-BAL-QUID-003-011 the temperature of reanalysis product has been compared with all available measurements across the Baltic for period of 1993-2024. The bias showed high temporal variability especially at surface where deviations from observations were in range of ± 0.2 °C. The average bias remained within range ± 0.3 °C at all depths. Largest positive bias was shown in subsurface layer between depths from 30 to 80 m. The average RMSE in that layer was also highest reaching 1.5 °C. The bias was negative in surface layer and deep layer, where mean bias was in range of -0.25 to -0.07 °C. The RMSE of surface and bottom layer was 0.9 °C and 0.3 °C, respectively.

The linear trends/slopes/fits and their 95% confidence intervals (Figure 1) have been calculated using last square regression method described in Draper and Smith (1998) eq. 1.2.9 and 1.4.8.

Scientific evaluation can be found in the CMEMS Ocean State Report (Guinehut et al., 2016; Mulet et al., 2018).

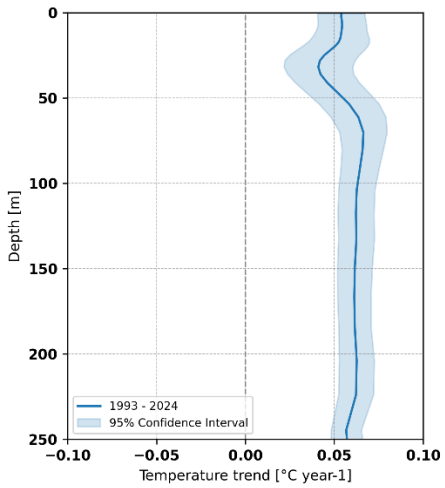

Figure 1. The vertical profile of subsurface horizontally averaged temperature trend for the period of 1993-2024. The temperature from Copernicus Marine Service regional reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 have been averaged over the Baltic Sea domain (13 °E - 31 °E; 53 °N - 66 °N; excluding the Skagerrak strait). The shaded area shows 95% confidence interval of calculated trend values.

3) REFERENCES

Draper, N. R., & Smith, H. (1998). Applied Regression Analysis. Wiley Series in Probability and Statistics. doi:10.1002/9781118625590

Guinehut et al., 2016: Surface and subsurface salinity. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

Mulet, S., Nardelli, BB., Good, S., Pisano, A., Greiner, E., Monier, M., Autret, E., Axell, L., Boberg, F., Ciliberti, S., Drévillon, M., Droghei, R., Embury, O., Gourrion, J., Høyer, J., Juza, M., Kennedy, J., Lemieux-Dudon, B., Peneva, E., Reid, R., Simoncelli, S., Storto, A., Tinker, J., von Schuckmann, K., Wakelin, S.L., 2018. Ocean temperature and salinity. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, s5–s13, DOI:10.1080/1755876X.2018.1489208